# A Scalable Multimodal Framework for Text–Image Coordination in User-Generated Tourism Content

# [Extended abstract]

Fariba Noorighomesheh[1,*], Aris Anagnostopoulos[1] and Ioannis Chatzigiannakis[1]

[1]Department of Computer, Control and Management Engineering (DIAG), Sapienza University of Rome, Rome, Italy

**Abstract**

Tourism platforms increasingly rely on user-generated content that combines long-form narratives with images. When the two modalities are weakly coordinated, they can introduce noise into search, recommendation, and content-understanding pipelines; however, large-scale assessment is difficult because real-world content is noisy, multilingual, and rarely accompanied by alignment labels. We present MMCEF, a modular, label-free framework for evaluating text–image coordination. The pipeline combines optional translation, text summarisation, vision–language captioning, and CLIP-based cross-modal scoring, while using a summarisation-based pseudo-reference only for internal configuration selection. Independent validation is provided through deliberately mismatched controls and human judgements. On KuriU, mean similarity decreases from 0.8620 to 0.4183 for the selected BERT-based text representation and from 0.7712 to 0.4251 for LLaVA captions after in-domain mismatching. A 100-pair human validation shows substantial agreement ($\alpha = 0.80$) and strong association with the proposed score ($r = 0.81$, $\rho = 0.85$). A proof-of-concept recommender experiment further shows that the score can act as an auxiliary post-hoc reranking signal, although the gains are not uniform and do not establish a causal effect on user preference.

**Keywords**

multimodal information retrieval, text–image coordination, user-generated content, recommender systems, label-free evaluation, tourism

## 1. Introduction and Motivation

User-generated multimodal content is central to modern tourism platforms. Travellers describe experiences through long-form reviews, travel narratives, and photographs, and these signals are subsequently reused by search, recommendation, ranking, and content-understanding systems. The information is valuable precisely because it is rich and subjective, but the same characteristics make it difficult to process reliably: text can be long, multilingual, informal, or redundant, while images may be weakly related to the accompanying narrative or may depict only a small part of the described experience. Such inconsistencies can become a source of noise for downstream information-access systems.

We focus on *text–image coordination*: the extent to which the textual and visual components of an item convey a coherent semantic account. This notion goes beyond object-level matching. A photo may contain plausible objects and still fail to support the narrative, context, or overall meaning expressed by the user. Existing vision–language encoders such as CLIP [1] provide strong cross-modal representations, while recent vision–language models can generate detailed image captions [3]. Nevertheless, real user-generated content differs substantially from the short, curated image–caption pairs commonly used in benchmark evaluation. In particular, supervised alignment evaluation is hard when no manually annotated ground truth exists at scale.

The goal of MMCEF (Multimodal Coordination Evaluation Framework) is therefore not to train a new foundation model, but to provide a modular post-hoc evaluation layer for noisy multimodal content. The framework combines existing pretrained components to construct multiple semantic

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*Corresponding author.

noorighomesheh.1958130@studenti.uniroma1.it (F. Noorighomesheh); aris@diag.uniroma1.it (A. Anagnostopoulos); ichatz@diag.uniroma1.it (I. Chatzigiannakis)



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views of an item and to compare them in a common evaluation protocol. Its design separates *internal configuration selection* from *external validation*: automatically generated summaries can support relative comparison among candidate configurations, but they are not treated as independent evidence of semantic correctness. Validation instead relies on controlled mismatching and human judgements.

This perspective is relevant to Information Retrieval and Recommender Systems in two ways. First, coordination can be used as a content-quality signal when indexing or ranking multimodal items. Second, it can complement collaborative signals when a recommendation pipeline must operate over noisy user-generated content. Recent recommender-system research has similarly highlighted the importance of modelling the reliability and consistency of multimodal user-generated information [5]. Our downstream experiment therefore treats the coordination score as an auxiliary reranking feature rather than as a direct model of user preference.

## 2. MMCEF: A Label-Free Coordination Framework

MMCEF processes textual and visual content through a sequence of modular stages. For multilingual inputs, text can first be translated into English with MarianMT [4]. The content-generation stage then creates complementary textual representations: summaries of the original narrative and captions derived from the associated images. In our experiments, BART, BERT-based summarisation, and T5 are compared for text condensation, while BLIP, Moondream, and LLaVA are compared for image captioning. The selected configuration is not fixed by design; the framework is intended to allow alternative pretrained components to be substituted as they become available.

CLIP is used as the primary cross-modal encoder because its image and text encoders map both modalities into a shared representation space [1]. Long textual descriptions exceed CLIP's default token limit, so they are processed in chunks and combined using alternative aggregation functions, including mean, maximum, weighted mean, geometric mean, and median. SBERT [2] is evaluated separately as a complementary text-only comparator for long textual representations; it is not used as a fused replacement for the visual branch of CLIP.

Because no manual alignment labels are available for the large corpora, the best-performing summarisation representation is used as a *pseudo-reference* for relative configuration comparison. This pseudo-reference is restricted to internal selection: error and correlation measures computed against it indicate consistency among candidate configurations rather than absolute semantic correctness. This distinction avoids using the same automatically generated signal as both the target and the final proof of validity.

We evaluate the framework on two tourism datasets. KuriU contains 16,368 narrative posts associated with 79,362 images; posts average 1,426 characters and are predominantly Italian. The TripAdvisor collection contains 5,445 English review–image pairs with an average text length of 423 characters. Two deliberately different mismatch conditions are used. *Shuffled KuriU* is an in-domain negative control in which each text is paired with a different image from the same KuriU corpus, preserving the image domain while destroying the original item-level association. *Shuffled TripAdvisor* pairs TripAdvisor text with unrelated Unsplash images and is therefore an external stress test that combines semantic mismatch with an image-domain shift.

## 3. Experimental Evidence

The configuration comparison selects LLaVA as the primary caption generator: its average caption–text similarity is 0.7712, compared with 0.7411 for BLIP and 0.7536 for Moondream. For the text representations, the BERT-based summary yields the strongest internal similarity among the compared summarisation alternatives. More importantly, the negative controls test whether high scores persist when the semantic relationship is deliberately broken.

Table 1 shows a clear decrease under deliberate mismatch. For KuriU, the average similarity drops from 0.8620 to 0.4183 for the BERT-based text representation and from 0.7712 to 0.4251 for LLaVA captions.

**Table 1**
Selected coordination results. KuriU uses an in-domain negative control; the TripAdvisor condition is a stronger cross-dataset stress test that also changes the image source.

| Evaluation | Original | Mismatched | Role |
|---|---|---|---|
| KuriU, BERT-based text representation | 0.8620 | 0.4183 | In-domain negative control |
| KuriU, LLaVA-caption representation | 0.7712 | 0.4251 | In-domain negative control |
| TripAdvisor, fixed selected configuration | 0.559 | 0.228 | External stress test |

**Table 2**
Selected NDCG@10 results from coordination-aware reranking. Relative changes are reported with respect to the MF baseline for the same interaction-density condition.

| Min. interactions | Users | Best setting | NDCG@10 | ΔNDCG@10 |
|---|---|---|---|---|
| 3 / 3 | 3428 | MF only | 0.0154 | 0.0% |
| 5 / 5 | 1534 | Mix, $\alpha = 0.1$ | 0.0198 | +4.5% |
| 8 / 8 | 550 | Additive, $\alpha = 0.3$ | 0.0442 | +9.2% |
| 10 / 10 | 321 | Mix, $\alpha = 0.1$ | 0.0701 | +2.2% |

Because both sides of the KuriU control remain within the same image domain, this experiment directly tests sensitivity to item-level reassignment. On TripAdvisor, the fixed BERT-summary/LLaVA-caption configuration yields a mean similarity of 0.559 on the original pairs and 0.228 on the shuffled condition. This latter drop should be interpreted as sensitivity under a stronger distributional perturbation, because the shuffled images come from Unsplash and thus introduce both semantic mismatch and image-domain shift.

We also perform an independent human validation on 100 KuriU stimuli, balanced between 50 original and 50 deliberately mismatched pairs. Participants rated semantic coherence on a five-point Likert scale without access to model-generated scores. Inter-annotator agreement is substantial (Krippendorff's $\alpha = 0.80$). The mean human ratings are strongly associated with the automatically generated coordination score (Pearson $r = 0.81$, $p < 10^{-23}$; Spearman $\rho = 0.85$, $p < 10^{-27}$). These results provide external evidence that the score reflects human-perceived semantic coherence within the evaluated sample. They should not, however, be interpreted as a fine-grained calibration result for all naturally matched items or as evidence about user preference.

## 4. Coordination-Aware Recommendation

To connect content coordination with downstream information access, we conduct a proof-of-concept recommendation experiment on KuriU implicit-feedback data. A matrix-factorisation (MF) model produces the initial candidate ranking. The coordination score $C(i)$ is then introduced only at the reranking stage through two simple strategies: an additive formulation, $s_{\text{final}}(u, i) = s_{\text{MF}}(u, i) + \alpha C(i)$, and a convex combination of normalized MF and coordination scores. We evaluate several interaction-density regimes and small coordination weights, using Precision@5, NDCG@10, and MAP.

Table 2 summarizes the NDCG@10 results. The sparsest 3/3 condition does not benefit from reranking, while gains appear at intermediate and denser regimes: +4.5% at 5/5, +9.2% at 8/8, and +2.2% at 10/10. The effect is not uniform across all ranking metrics; for example, MAP decreases slightly in the 10/10 condition. We therefore interpret the experiment as evidence that semantic coordination can provide useful auxiliary information for offline reranking in some regimes, rather than as proof that users causally prefer higher-coordination items. Historical interactions may already reflect exposure, popularity, creator effort, presentation quality, or other factors correlated with semantic coherence. Establishing causality would require prospective or online evaluation.

## 5. Discussion and Conclusion

This extended abstract presents ongoing work on scalable text–image coordination for noisy tourism UGC. The main contribution is a label-free, modular evaluation protocol that separates automatic configuration selection from independent validation and that can operate over long-form, multilingual content. The results indicate that the selected score reacts strongly to controlled mismatching, aligns with human coherence judgements in the evaluated sample, and can be used as a lightweight post-hoc signal in a recommender-system pipeline.

The study has several limitations. Both empirical datasets are tourism datasets, so transfer beyond tourism remains to be established. The human study is intentionally small and is used for perceptual validation rather than population-level inference. The TripAdvisor shuffled condition is a stress test rather than a pure in-domain mismatch control, and the recommender experiment is observational and offline. These limitations motivate broader evaluation against retrieval- and entailment-based alignment metrics, newer vision–language models, additional domains, and online recommendation settings.

## Declaration on Generative AI

During the preparation of this work, the authors used OpenAI ChatGPT to assist with drafting content and abstract generation by condensing and restructuring material from the authors' existing manuscript into an extended-abstract format, as well as paraphrasing and rewording, improving writing style, grammar and spelling checking, and formatting assistance. No scientific results, data, analyses, or conclusions were generated by the tool. The authors reviewed and edited the resulting content and take full responsibility for the publication's content.

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